

1(a)	any three from: 1 osmoreceptors detect (decrease) or osmoreceptors, shrink / decrease in volume ; 2 <i>ref. to</i> hypothalamus ; 3 hypothalamus produces (more) ADH ; A neurosecretory cells produce ADH R secretion 4 <u>posterior pituitary</u> releases (more) ADH (into blood) ; A secretes R produces	3
1(b)(i)	second(ary) messenger ;	1
1(b)(ii)	adds aquaporins into, cell surface / luminal, membrane ; (so) more permeable to water ;	2
1(b)(iii)	arrow from lumen into cell and arrow from cell into capillary or one arrow from lumen through cell and into capillary ;	1

Question	Answer	Marks
1(c)	<i>any two from:</i> 1 no / less, water reabsorbed or no / less, water absorbed into, capillary / blood ; ↓ tissue fluid 2 water potential of blood, (remains) low / lower than set point ; 3 large(r) volume of urine / dilute urine / low(er) concentration of urine ; ↓ amount / more water excreted / more urine 4 AVP ; e.g. feelings of thirst / dehydration / dizzy / fatigue / ADH release continues / decreased blood volume	2